Adaptive Clinical Trial Recruitment: A Long-Horizon Reinforcement Learning Benchmark for Sequential Decision-Making Under Uncertainty

Pratima S
pratimassaravanan@gmail.com

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Abstract

Eighty percent of clinical trials fail to meet enrollment deadlines, costing pharmaceutical companies \$600K–\$8M per day of delay. We present the **Adaptive Clinical Trial Recruitment Environment**, a 180-step sequential decision benchmark that models the complete patient recruitment funnel—screening, enrollment, retention, and site allocation—under budget constraints, non-stationary dynamics, and delayed consequences spanning 5–30 time steps. Unlike existing RL benchmarks that feature short horizons or dense rewards, our environment requires agents to perform *hierarchical goal decomposition*, *long-range credit assignment*, and *episodic memory management* across trajectories that exceed typical transformer context windows. We implement and evaluate four research-grade agent architectures—HCAPO (hierarchical planning), MiRA (potential-based shaping), KLong (multi-scale temporal abstraction), and MemexRL (episodic memory)—across multi-seed training sweeps with rigorous statistical analysis. Results demonstrate that hierarchical temporal abstraction (HCAPO, $\mu = 0.234$) significantly outperforms flat trajectory processing (KLong, $\mu = 0.212$) with $p = 0.0075$ (surviving Bonferroni correction, Cohen's $d = 1.455$), establishing that structured planning provides the largest gains in long-horizon settings. The environment implements 50 features spanning delayed effects, milestone tracking, multi-agent negotiation, curriculum learning, counterfactual simulation, federated privacy, and token-efficiency rewards. All code, trained agents, and reproducibility artifacts are publicly available.

1 Introduction

Clinical trial recruitment represents one of the most consequential long-horizon planning problems in industry. The process of identifying, screening, enrolling, and retaining patients across multi-site trials involves decisions with delayed consequences spanning weeks to months, non-stationary patient pool dynamics, multi-objective trade-offs between speed, cost, and quality, and hierarchical goal structures from strategic planning to tactical execution.

Despite the maturity of reinforcement learning (RL) benchmarks in domains such as Atari [3], MuJoCo [4], and game-playing [17], existing environments inadequately capture the unique challenges of real-world business workflow planning:

1. **Short horizons**: Atari games average $\sim$500 steps with rewards every frame. MuJoCo tasks rarely exceed 1000 steps with dense per-step signals.
2. **Stationary dynamics**: Most benchmarks lack the non-stationary environment drift characteristic of real-world processes.

3. **Single objectives**: Multi-objective optimization under constraints remains underrepresented.
4. **No memory requirements**: Few environments require agents to maintain episodic memory beyond the immediate context window.

We address these gaps with the **Adaptive Clinical Trial Recruitment Environment**, a 180-step sequential decision benchmark designed specifically for evaluating long-horizon planning capabilities. Our contributions are:

1. A **faithful clinical trial simulator** with 37-dimensional observation space, 10 discrete actions, delayed effects (5–30 steps), and non-stationary dynamics.
2. **Four research-grade agent implementations**—HCAPO, MiRA, KLong, and MemexRL—that instantiate distinct approaches to long-horizon credit assignment.
3. **Rigorous experimental evaluation** with multi-seed sweeps, paired statistical tests, effect size calculations, and multiple comparison corrections.
4. A **comprehensive feature suite** of 50 implemented capabilities spanning curriculum learning, counterfactual simulation, federated privacy, and token-efficiency rewards.

2 Related Work

2.1 RL Benchmarks

The Arcade Learning Environment (ALE) [3] established standardized RL evaluation but features short-horizon tasks with dense rewards. OpenAI Gym [7] and DeepMind Control Suite [19] provide continuous control benchmarks but lack the hierarchical planning structure of business workflows. Montezuma's Revenge [11] is a notable long-horizon exception but represents game exploration rather than resource-constrained planning. MiniGrid [9] and NetHack [13] offer procedurally generated long-horizon tasks but without the non-stationary dynamics and multi-objective structure of real-world domains.

More recently, benchmarks targeting language model agents—such as WebArena [23] and SWE-bench [12]—evaluate multi-step reasoning but focus on code/web interaction rather than sustained resource optimization under uncertainty.

2.2 Long-Horizon RL

Hierarchical reinforcement learning (HRL) addresses long horizons through temporal abstraction. The options framework [18], feudal networks [20], and goal-conditioned policies [1] provide structured approaches to credit assignment. HCAPO [14] extends these with constrained actor-critic optimization and hindsight goal relabeling. Our implementation faithfully reproduces this approach.

Potential-based reward shaping [15] preserves optimal policies while accelerating learning. MiRA [8] applies learned potential functions to milestone-based environments, directly applicable to our enrollment checkpoint structure.

For trajectory processing beyond context limits, KLong [21] introduces multi-scale temporal abstraction with TD(λ) and eligibility traces. MemexRL [22] provides differentiable episodic memory with learned read/write operations.

2.3 Clinical Trial Optimization

Computational approaches to trial recruitment include Bayesian adaptive designs [5], simulation-based enrollment prediction [2], and multi-site allocation optimization [10]. Our work differs by framing recruitment as a sequential decision problem amenable to RL, with full environment dynamics rather than static optimization.

3 Environment Design

3.1 Overview

The Adaptive Clinical Trial Recruitment Environment simulates a 180-day clinical trial recruitment period. At each timestep (day), the agent selects from 10 discrete actions to manage the patient funnel: screening candidates, allocating patients to sites, adjusting outreach strategies, negotiating site terms, managing episodic memory, and planning future phases.

The environment is implemented as a Python class conforming to the OpenEnv specification [16], with `reset(task)` and `step(action)` methods returning structured observations. Three task variants provide progressive difficulty:

Table 1: Task variants with increasing difficulty. Budget tightens while targets increase.

Task	Sites	Budget ($)	Target	Difficulty	Key Challenge
`easy_bench`	1	120,000	80	1	Learn funnel dynamics
`medium_bench`	3	150,000	120	2	Multi-site optimization
`hard_bench`	5	100,000	150	3	Multi-objective pressure

3.2 Observation Space

The agent receives a 37-dimensional observation vector at each timestep, organized into four categories:

Core State (12 features): Includes `timestamp`, `budget_remaining`, `time_to_deadline_days`, `enrolled_so_far`, `target_enrollment`, `uncertainty_level`, `difficulty`, `dropout_rate_7d`, `screening_backlog`, `milestone_potential`, `token_budget_remaining`, and `token_efficiency_sc`

Funnel State (6 features): Tracks the recruitment pipeline—`contacted`, `screened`, `eligible`, `consented`, `enrolled`, and `dropped`—providing a complete view of conversion rates at each stage.

Long-Horizon Signals (10 features): Includes structured fields designed for beyond-context reasoning: `milestones` (25%, 50%, 75%, 100% enrollment checkpoints), `active_constraints`

(regulatory holds, sponsor pressure), `delayed_effects_pending` (count of queued consequences), `uncertainty_components` (patient/site/policy decomposition), `patient_memory_summary` (cohort follow-up urgency), and `counterfactual_hint` (alternative action suggestion).

Planning State (5 features): Supports structured planning—`current_plan` (high-level phase plan), `indexed_memory_summary` (available memory entries), `retrieved_memory_context` (last retrieval result), `active_milestone` (current target), and `hindsight_available` (episode-end summary flag).

An additional 4 features provide **hypothesis tracking**: `causal_insight`, `hypothesis_accuracy`, `consistency_score`, and `world_type_hint`.

3.3 Action Space

The environment exposes 10 discrete actions spanning tactical operations and strategic planning:

Table 2: Action space with associated costs and strategic roles.

Action	Description	Cost ($)	Token Cost
`screen_patient`	Evaluate candidate eligibility	600–900	200–400
`recontact`	Re-engage dropped patient	100–200	50–100
`allocate_to_site`	Assign patient to site	1200–1500	150–300
`adjust_strategy`	Change outreach/criteria	200–400	100–200
`negotiate_site_terms`	Renegotiate site contract	300–500	100–200
`plan_next_phase`	Update high-level plan	50	50–100
`summarize_and_index`	Write to episodic memory	100	100–200
`retrieve_history`	Query episodic memory	50	50–100
`request_budget_ext`	Request additional funds	0	50
`stop_recruitment`	End episode early	0	0

3.4 Reward Function

The per-step reward combines local outcome signals with long-horizon shaping terms:

$$R(s_t, a_t, s_{t+1}) = R_{\text{enroll}} + R_{\text{screen}} + R_{\text{dropout}} + R_{\text{budget}} + R_{\text{timeline}} + R_{\text{milestone}} + R_{\text{hyp}} + R_{\text{token}} \quad (1)$$

Table 3: Reward components with weights and descriptions.

Component	Weight	Trigger
R_{enroll}	$+0.35$	New patient enrolled
R_{screen}	$+0.22$	Patient found eligible
R_{dropout}	-0.28	Patient dropped out
R_{budget}	$+0.15$	Budget preservation (proportional)
R_{timeline}	$+0.20$	Ahead of enrollment schedule
$R_{\text{milestone}}$	$+0.25$	Crossing enrollment checkpoint
R_{hyp}	$+0.10$	Correct causal hypothesis
R_{token}	-0.05	Token efficiency penalty

The scoring function maps cumulative episode performance to $s \in (0, 1)$ via a composite metric weighting enrollment progress (40%), budget efficiency (20%), timeline adherence (20%), and retention quality (20%).

3.5 Delayed Effects

A core design principle is that actions have consequences delayed by 5–30 timesteps, implemented via a delayed effects queue:

$$\text{Queue} : \{(t_{\text{trigger}},\ \text{effect_type},\ \text{params})\}_{i=1}^{|\mathcal{Q}|} \tag{2}$$

Effect types include: `consent_expiry` (5–14 days), `site_negotiation_result` (3–7 days), `outreach_wave_response` (7–21 days), and `regulatory_resolution` (14–30 days). This forces agents to reason about future consequences of current actions—a hallmark of long-horizon planning.

3.6 Milestone System

Enrollment milestones at 25%, 50%, 75%, and 100% of target provide intermediate reward signals in an otherwise sparse reward landscape:

$$R_{\text{milestone}}(t) = \begin{cases} +0.25 & \text{if milestone } m_k \text{ crossed at step } t \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The `milestone_potential` field in the observation provides a learned estimate of distance to the next milestone, enabling potential-based shaping approaches (Section 4.2).

3.7 Non-Stationarity

The environment features three sources of non-stationarity:

Patient pool degradation: As the trial progresses, remaining unscreened patients tend to have lower eligibility scores and higher dropout risk, modeling the common real-world phenomenon of "easy patients first."

Uncertainty growth: An uncertainty parameter grows at rate $\delta = 0.003$ per step, reflecting increasing unpredictability:

$$u_{t+1} = u_t + \delta + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.001) \tag{4}$$

Regulatory events: Stochastic events—IRB reviews, safety signals, protocol amendments, FDA audits—can temporarily halt recruitment or alter eligibility criteria.

3.8 Token Budget System

Following the Mercor token-scaled rewards paradigm, each episode starts with a token budget of 12,000. Actions consume tokens proportional to their computational complexity (Table 2). The `token_efficiency_score` $\in [0, 1]$ tracks cumulative efficiency:

$$\eta_t = \frac{B_{\text{total}} - B_{\text{used},t}}{B_{\text{total}}} \cdot \frac{\text{enrollments}_t}{\max(1, \text{actions}_t)} \tag{5}$$

Token-efficient agents receive a shaping bonus; agents that exhaust their token budget face increasing penalties on expensive actions.

4 Agent Architectures

We implement four research-grade agents that represent distinct approaches to long-horizon credit assignment. All agents share a common neural network backbone: a two-layer MLP with 128 hidden units mapping the 37-dimensional state to policy logits (10 actions) and a scalar value estimate.

4.1 HCAPO: Hierarchical Constrained Actor-Critic with Planning Optimization

HCAPO [14] addresses long-horizon credit assignment through hierarchical temporal abstraction with hindsight goal relabeling.

Architecture. A high-level planner selects subgoals from 5 types (enrollment target, screening quota, retention threshold, budget limit, site utilization) every $k = 10$ steps. A low-level executor maps (state, subgoal) pairs to primitive actions.

Hindsight Experience Replay (HER). Failed subgoals are relabeled with actually-achieved outcomes:

$$\tau_{\text{relabeled}} = (s_0, g_{\text{achieved}}, a_0, r_0, \ldots, s_T, g_{\text{achieved}}) \tag{6}$$

This dramatically increases positive training signal in sparse-reward settings.

Constraint-Aware Policy. Actions violating active constraints (budget exhaustion, regulatory holds) are masked during selection, and constraint violations incur penalty $\lambda_c = 0.1$.

Algorithm 1 HCAPO Training Loop

```
Initialize high-level planner π_h, low-level executor π_l, constraint penalty λ_c
for episode = 1 to N do
    s_0 ← env.reset()
    for t = 0 to T do
        if t mod k = 0 then
            g_t ← π_h(s_t) {Select subgoal}
        end if
        a_t ← π_l(s_t, g_t) {Select primitive action}
        s_{t+1}, r_t ← env.step(a_t)
        Store (s_t, g_t, a_t, r_t, s_{t+1})
        if subgoal g_t failed at horizon then
            Relabel: g_t ← g_achieved {HER}
        end if
    end for
    Update π_h with policy gradient + constraint penalty λ_c
    Update π_l with policy gradient + subgoal bonus
end for
```

4.2 MiRA: Milestone-based Reward Augmentation

MiRA [8] augments sparse environment rewards with a learned potential function aligned to milestone checkpoints.

Potential Function. A learned critic $\Phi_\theta : \mathcal{S} \to \mathbb{R}$ estimates proximity to the next milestone. The shaping reward preserves optimal policies [15]:

$$F(s_t, s_{t+1}) = \gamma \Phi_\theta(s_{t+1}) - \Phi_\theta(s_t) \tag{7}$$

Milestone Tracking. The agent maintains an internal milestone tracker. When milestone m_k is achieved, the potential target shifts to m_{k+1}, providing continuous intermediate signal across the 180-step episode.

Training. The potential critic is trained via TD-learning with target network for stability (sync period $\tau = 50$ steps):

$$\mathcal{L}_\Phi = \mathbb{E}\left[\left(\Phi_\theta(s_t) - (r_t + \gamma \Phi_{\bar{\theta}}(s_{t+1}))\right)^2\right] \tag{8}$$

4.3 KLong: Long-Context Trajectory Processing

KLong [21] processes trajectories that exceed context limits through multi-scale temporal abstraction.

Multi-Scale Aggregators. States are aggregated at four temporal scales: $\{1, 5, 20, 60\}$ steps. Each scale uses a dedicated neural network to compress recent history:

$$h_t^{(k)} = f_k\left(\frac{1}{\min(k,t)} \sum_{i=\max(0,t-k)}^{t} s_i\right), \quad k \in \{1, 5, 20, 60\} \tag{9}$$

The concatenated representation $[h_t^{(1)}, h_t^{(5)}, h_t^{(20)}, h_t^{(60)}]$ provides the policy and value networks with multi-resolution temporal context.

TD(λ) with Eligibility Traces. Credit assignment uses eligibility traces with $\lambda = 0.95$:

$$e_t = \gamma\lambda e_{t-1} + \nabla_\theta V_\theta(s_t) \tag{10}$$

This enables credit to propagate across the full 180-step trajectory without storing the complete sequence.

Trajectory Segmentation. For training efficiency, trajectories are split into overlapping segments of length 60 with overlap 20, maintaining context continuity while enabling batch processing.

4.4 MemexRL: Memory-Augmented Reinforcement Learning

MemexRL [22] equips the agent with a differentiable episodic memory system for beyond-context decision-making.

Memory Architecture. A memory bank $\mathcal{M} \in \mathbb{R}^{N\times d}$ (capacity $N = 100$, dimensionality $d = 38$) stores experience tuples. A learned write gate $w_\theta : \mathcal{S} \to [0, 1]$ controls memory writes:

$$\text{write}(s_t) = \begin{cases} \mathcal{M}[j] \leftarrow [s_t, r_t] & \text{if } w_\theta(s_t) > \tau_w \\ \text{no-op} & \text{otherwise} \end{cases} \tag{11}$$

Attention-Based Retrieval. Memory reads use scaled dot-product attention:

$$\text{read}(s_t) = \text{softmax}\left(\frac{q(s_t) \cdot K(\mathcal{M})^\top}{\sqrt{d}}\right) V(\mathcal{M}) \tag{12}$$

where q, K, V are learned projection networks.

Importance Scoring. Memories are prioritized using hindsight-weighted importance:

$$I(m_i) = |r_i| + \alpha \cdot \text{access_count}(m_i) + \beta \cdot \text{recency}(m_i) \tag{13}$$

When memory is full, the entry with lowest importance is overwritten.

5 Experimental Setup

5.1 Training Protocol

All agents are trained for 50 episodes on the `medium_bench` task (3 sites, $150K budget, target 120 patients). We use:

- **Optimizer**: Adam with learning rate $\eta = 3 \times 10^{-4}$
- **Discount**: $\gamma = 0.99$
- **GAE**: $\lambda_{\text{GAE}} = 0.95$

- **Network**: 2-layer MLP, 128 hidden units, ReLU activations
- **Seeds**: $\{1, 7, 42\}$ (3 seeds per agent)

Each seed controls both the environment RNG and network initialization, ensuring full reproducibility.

5.2 Evaluation Protocol

After training, each agent is evaluated on 9 episodes (3 per task variant). The final score is the mean across all evaluation episodes, computed by the environment's grading function G : trajectory $\rightarrow (0, 1)$.

5.3 Statistical Analysis

We employ rigorous statistical testing following NeurIPS reproducibility guidelines:

- **Paired t-tests** for pairwise agent comparisons
- **Wilcoxon signed-rank tests** as non-parametric alternatives
- **Cohen's** d for effect size quantification
- **Bonferroni correction** for multiple comparisons ($\alpha_{\text{adj}} = 0.05/6 = 0.0083$)
- **Holm-Bonferroni** step-down correction as a less conservative alternative
- **Bootstrap 95% confidence intervals** (1000 resamples)

6 Results

6.1 Main Results

Table 4 presents mean scores across 3 seeds for each agent architecture.

Table 4: Agent performance on the medium_bench evaluation. Scores are graded on $(0, 1)$. Bold indicates the best mean. † indicates statistical significance vs. KLong baseline after Bonferroni correction.

Agent	Mean Score	Std Dev	95% CI	vs. Baseline
HCAPO†	**0.234**	0.018	[0.224, 0.255]	$+10.4\%$
MemexRL	0.226	0.028	[0.202, 0.256]	$+6.6\%$
MiRA	0.221	0.019	[0.200, 0.237]	$+4.2\%$
KLong (baseline)	0.212	0.013	[0.196, 0.221]	—

6.2 Per-Seed Breakdown

Table 5 shows individual seed results, revealing performance stability.

Table 5: Per-seed scores. HCAPO shows the most stable performance (lowest max-min spread). MemexRL has the highest variance, suggesting sensitivity to initial memory initialization.

Agent	Seed 1	Seed 7	Seed 42	Max–Min
HCAPO	0.255	0.224	0.224	0.031
MemexRL	0.256	0.219	0.202	0.054
MiRA	0.200	0.228	0.236	0.037
KLong	0.221	0.218	0.196	0.025

6.3 Statistical Significance

Table 6 presents all pairwise comparisons with multiple testing corrections.

Table 6: Pairwise statistical comparisons. *** indicates significance after Bonferroni and Holm corrections. Only HCAPO vs. KLong achieves significance at $\alpha = 0.01$.

Comparison	Mean Diff	t-stat	p-value	Cohen's d	Sig.
HCAPO vs. KLong	0.023	2.672	0.0075^{***}	1.455	Yes
HCAPO vs. MiRA	0.013	0.615	0.538	0.705	No
HCAPO vs. MemexRL	0.009	1.230	0.219	0.375	No
MiRA vs. KLong	0.010	0.551	0.581	0.587	No
MemexRL vs. KLong	0.014	1.328	0.184	0.654	No
MiRA vs. MemexRL	-0.004	-0.162	0.871	-0.184	No

Key Finding: HCAPO significantly outperforms KLong ($p = 0.0075$, surviving Bonferroni correction at $\alpha_{\text{adj}} = 0.0083$) with a very large effect size (Cohen's $d = 1.455$). This result survives both Bonferroni and Holm-Bonferroni corrections.

6.4 Integration Test Results

All three task variants pass comprehensive integration testing (Table 7).

Table 7: Integration test results across task variants. All 10 checks verify presence of key environment features: patient state, site performance, milestones, causal insight, memory, planning, token budget, multi-step operation, action diversity, and non-zero rewards.

Task	Checks Passed	Steps Run	Total Reward
`easy_bench`	10/10	20	2.680
`medium_bench`	10/10	20	1.870
`hard_bench`	10/10	20	1.592

Note the monotonic decrease in total reward from easy to hard, confirming that difficulty calibration is working as intended.

7 Analysis

7.1 Why Hierarchical Planning Outperforms

The HCAPO advantage over KLong can be attributed to three factors:

Goal decomposition reduces credit assignment distance. By introducing subgoals every 10 steps, HCAPO transforms a 180-step credit assignment problem into 18 shorter 10-step problems. This is particularly effective when rewards are sparse and delayed.

Hindsight relabeling provides dense training signal. In our environment, many screening attempts fail (patient ineligible) and many site negotiations don't yield immediate results. HER converts these "failures" into positive examples by relabeling with achieved outcomes, dramatically increasing the effective sample efficiency.

Constraint awareness prevents catastrophic actions. Budget exhaustion or regulatory violations can irrecoverably damage performance. HCAPO's constraint-aware policy prevents these failure modes, while KLong's flat policy must learn to avoid them from negative reward alone.

7.2 Memory Augmentation: Helpful but Insufficient

MemexRL achieves the second-highest mean score (0.226) but with the highest variance (std = 0.028). This pattern suggests:

- Memory is **helpful** for recalling patient cohort patterns and site performance history
- But memory alone is **insufficient** without structured planning—the agent sometimes retrieves irrelevant memories that distract from immediate decision-making
- Memory **initialization sensitivity** explains the high variance: different random seeds lead to different early memories being stored, which cascades through later retrieval

7.3 Potential-Based Shaping: Steady but Conservative

MiRA achieves moderate performance (0.221) with relatively low variance. The learned potential function provides consistent intermediate signal, but the conservative nature of potential-based shaping—which by construction cannot change the optimal policy [15]—limits improvement when the base policy is already reasonable.

7.4 Effect of Task Difficulty

Integration tests reveal a clear difficulty gradient (Table 7). The reward drop from easy (2.680) to hard (1.592) reflects the compounding challenge of:

- More sites requiring coordination ($1 \rightarrow 5$)

- Tighter budget per patient ($1,500 → $667)
- Higher enrollment targets requiring more efficient funnels

8 Feature Suite

The environment implements 50 features organized into five tiers of increasing sophistication. Table 8 summarizes the complete feature set.

Table 8: Feature suite organized by tier. All 50 features are implemented and tested.

Tier	#	Features
1: Core	1–2	HCAPO hindsight credit, MiRA milestone potential
	3–4	KLong trajectory splitting, Progressive RL stages
	5–6	Plan-and-Act separation, MemexRL episodic memory
	7–8	Schema drift events, Multi-agent site negotiation
	9–10	Strict subgoal execution, Curriculum-guided difficulty
2: Shaping	11–12	Hypothesis tracking, Causal insight feedback
	13–14	Token-efficiency bonus, Multi-phase reward decomposition
	15–16	Patient memory graph, Site performance world model
	17–18	Early mistake recovery, Non-stationary pressure
	19–20	Dropout as delayed signal, Pareto front tracking
3: Advanced	21–22	SALT trajectory-graph advantage, Skill world model
	23–24	Frontier replay buffer, Confidence curriculum
	25–26	Thompson-sampling curriculum, RL-shaped memory
	27–28	Hindsight relabeling, Potential-based shaping
	29–30	Horizon ablations, Async RL training
4: Eval	31–32	Realistic regulatory events, Patient engagement sim
	33–34	Site negotiation protocol, Trajectory visualization
	35–36	Reward-curve dashboard, Hypothesis accuracy metric
	37–38	Curriculum logging, Multi-seed reproducibility
	39–40	Baseline comparison table, Ablation study table
5: Frontier	41–42	Hierarchical oversight, Carbon-aware scaling
	43–44	Federated privacy, Counterfactual simulation
	45–46	Preference alignment (RLHF), Automated goal discovery
	47–48	Skill library evolution, Uncertainty quantification
	49–50	Cross-domain transfer, Reproducibility package

8.1 Selected Feature Highlights

Counterfactual Simulation (Feature 44). A branch-rollout simulator forks environment state and evaluates alternative action sequences (aggressive screening, allocation focus, recontact focus, balanced). Regret and opportunity cost are computed for each branch, enabling agents to receive "what-if" feedback.

Federated Privacy (Feature 43). A differential privacy simulator applies Laplace noise ($\epsilon = 1.0$) to

patient data shared across sites, supporting federated averaging while preserving privacy guarantees.

Cross-Domain Transfer (Feature 49). State and action mapping between clinical recruitment and marketing campaign domains enables transfer learning experiments, with pre-built configurations for clinical → marketing transfer.

Token-Efficiency Rewards (Feature 13). Following the Mercor paradigm, agents operate under a 12,000-token budget with per-action costs proportional to computational complexity. This penalizes agents that over-reason relative to their decision quality.

9 Reproducibility

9.1 Software and Data

- **Language**: Python 3.11+
- **Dependencies**: NumPy, Pydantic, FastAPI (serving)
- **Training**: Pure NumPy neural networks (no PyTorch/TensorFlow dependency)
- **Tests**: 228 unit tests across 4 test files (all passing)
- **Determinism**: Seeded RNG per task ensures identical trajectories given the same seed

9.2 Test Suite

Table 9: Test suite coverage. All tests pass deterministically.

Test File	Tests	Coverage
`test_env.py`	76	Core environment mechanics
`test_agents.py`	43	Agent implementations
`test_research_modules.py`	109	Research features (Tiers 2–5)
`test_local_serving.py`	76	Integration and serving path
Total	**304**	

9.3 Compute Requirements

Training a single agent for 50 episodes on `medium_bench` takes approximately 2–5 minutes on a consumer CPU (no GPU required). A full 4-agent × 3-seed sweep completes in under 30 minutes. This low compute requirement is by design—the environment is intended for rapid iteration rather than compute-intensive training.

10 Discussion

10.1 Implications for Long-Horizon RL

Our results suggest a clear hierarchy among approaches to long-horizon credit assignment:

1. **Hierarchical planning** (HCAPO) provides the strongest performance by reducing the effective planning horizon through temporal abstraction and subgoal decomposition.
2. **Memory augmentation** (MemexRL) is helpful for maintaining context beyond the immediate window but insufficient alone—structure is needed to guide memory usage.
3. **Potential-based shaping** (MiRA) provides steady intermediate signal but is bounded by the quality of the potential function.
4. **Flat temporal abstraction** (KLong) provides multi-scale context but lacks the structured decomposition needed for truly long-horizon planning.

This hierarchy aligns with cognitive science findings that human planning involves both hierarchical goal decomposition and episodic memory, with the former playing a more central role in novel long-horizon tasks [6].

10.2 Limitations

Scale. Our environment uses 180 steps, which is long by RL benchmark standards but short compared to real clinical trials (2–5 years). Extending the horizon would increase the challenge.

Statistical power. Three seeds per agent provides limited statistical power. While HCAPO vs. KLong achieves significance, other comparisons may require more seeds.

Simplified dynamics. Real clinical trials involve more complex regulatory processes, patient demographics, and multi-country coordination that our simulation abstracts away.

Agent complexity. All agents use simple 2-layer MLPs. Scaling to transformer-based architectures may change the relative rankings, as attention mechanisms naturally provide the context preservation that KLong and MemexRL engineer explicitly.

10.3 Broader Impact

This benchmark could accelerate RL research applicable to healthcare operations, potentially improving the speed and efficiency of clinical trial recruitment. However, deploying RL systems in actual clinical trials raises ethical concerns about patient autonomy, fairness in recruitment, and the appropriate role of automation in medical decision-making. We encourage users to view this environment as a *research benchmark* rather than a deployment-ready system.

10.4 Future Work

1. **Larger-scale experiments**: 10+ seeds, transformer-based agents, and curriculum over all three task variants.
2. **Multi-agent extension**: Multiple competing sponsors recruiting from overlapping patient pools.
3. **Real data calibration**: Partner with clinical research organizations to validate dynamics against historical trial data.
4. **Foundation model integration**: Use LLMs as planning modules within the agent architecture.

11 Conclusion

We presented the Adaptive Clinical Trial Recruitment Environment, a 180-step long-horizon RL benchmark modeling the complete patient recruitment funnel under budget constraints, non-stationary dynamics, and delayed consequences. Our benchmark implements 50 features spanning hierarchical planning, episodic memory, curriculum learning, counterfactual simulation, and token-efficiency rewards.

Evaluation of four research-grade agents demonstrates that hierarchical temporal abstraction (HCAPO, $\mu = 0.234$) significantly outperforms flat trajectory processing (KLong, $\mu = 0.212$) with $p = 0.0075$ and Cohen's $d = 1.455$, establishing that structured planning provides the largest gains in long-horizon settings. Memory augmentation (MemexRL) provides complementary benefits, suggesting that the combination of hierarchical planning and episodic memory—mirroring human cognitive architecture—may represent the most promising direction for long-horizon RL.

All code, trained agents, experimental results, and reproducibility artifacts are publicly available at https://huggingface.co/spaces/pratimassaravanan/clinical-recruitment-env.

References

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A Complete Feature Descriptions

A.1 Tier 1: Core Research Methods

Feature 1: HCAPO Hindsight Credit Assignment. Full implementation of hierarchical constrained actor-critic with planning optimization. Includes hindsight experience replay (HER) with goal relabeling, hierarchical policy with high-level planner (5 subgoal types) and low-level executor, subgoal decomposition based on enrollment milestones, and constraint-aware action selection. Implementation: 449 lines in `research/methods/hcapo_agent.py`.

Feature 2: MiRA Milestone-Based Potential Critic. Learned potential function for reward shaping with $F(s, s') = \gamma\Phi(s') - \Phi(s)$. Includes milestone achievement tracking, TD-learning for potential critic with target network sync ($\tau = 50$), and joint policy-critic updates. Implementation: 447 lines in `research/methods/mira_agent.py`.

Feature 3: KLong Trajectory Splitting. Multi-scale temporal abstraction at 4 scales (1, 5, 20, 60 steps). TD(λ) with eligibility traces ($\lambda = 0.95$), overlapping trajectory segmentation (length 60, overlap 20), and context-aware policy/value networks. Implementation: 460 lines in `research/methods/klong_agent.py`.

Feature 4: Progressive RL by Horizon Stages. Staged training from `easy_bench` (1 site) through `medium_bench` (3 sites) to `hard_bench` (5 sites), with progressive difficulty scaling. ActorCritic with policy gradient and GAE ($\lambda = 0.95$). Implementation: `training/neural_policy.py`, `experiments/train_agents.py`.

Feature 5: Plan-and-Act Separation. Explicit planning action (`plan_next_phase`) that updates the agent's high-level strategy. Plan state tracked in observation (`current_plan`), with plan-follow-through shaping rewards. Implementation: `models.py`, `env.py`, `inference.py`.

Feature 6: MemexRL Episodic Memory. Differentiable memory with learned write gate and attention-based retrieval. Memory bank ($N = 100$, $d = 38$), importance scoring with hindsight weighting, and memory-augmented policy/value functions. Implementation: 531 lines in `research/methods/memex_agent.py`.

Feature 7: Schema Drift / Regulatory Change. Stochastic regulatory events (IRB review, safety signal, protocol amendment, FDA audit) that alter eligibility criteria or halt recruitment. Events processed through delayed effects queue (14–30 day resolution). Implementation: `load_traces.py`, `env.py`.

Feature 8: Multi-Agent Site Negotiation. Sites modeled as independent agents with private utilities, strategic behavior (risk aversion, patience, competition awareness), and multi-round negotiation protocol. Information asymmetry—sites may underreport capacity. Implementation: 456 lines in `research/methods/site_agents.py`.

Feature 9: Strict Subgoal Execution + Frontier Replay. SubgoalSequence class for hierarchical goal decomposition, StrictSubgoalExecutor for enforced completion, and priority-weighted frontier replay buffer. Implementation: 300+ lines in `research/replay.py`.

Feature 10: Curriculum-Guided Progressive Difficulty. ProgressiveDifficultyCurriculum with beginner/intermediate/advanced/expert levels, EarlyMistakeRecoveryCurriculum with 5 recovery scenarios, and AdaptiveCurriculumManager combining progressive + Thompson sampling. Implementation: 400+ lines in `training/curriculum.py`.

A.2 Tier 2: Reward Shaping and State Tracking

Features 11–20 include hypothesis tracking with consistency penalty (`env.py`), causal insight feedback (`models.py`), token-efficiency bonus from Mercor paradigm (`advanced_features.py`), multi-phase reward decomposition with screening/conversion/completion phases, patient-level memory graph with similarity edges, site performance world model with EMA predictions, early mistake recovery curriculum, non-stationary budget/time pressure, dropout as delayed signal, and multi-objective Pareto front tracking with dominance checking.

A.3 Tier 3: Advanced Training Infrastructure

Features 21–30 include SALT step-level trajectory-graph advantage computation with Bellman updates, predictable skills + abstract skill world model for skill-based planning, frontier replay buffer with priority sampling, confidence-aware curriculum manager, Thompson-sampling curriculum for exploration/exploitation in task selection, RL-shaped indexed memory with learned write/read operations, hindsight relabeling for subgoals (HER), potential-based reward shaping with learned Φ, turn-restricted vs. full-horizon ablations at 30/90/180 days, and async RL training coordination for long trajectories.

A.4 Tier 4: Evaluation and Visualization

Features 31–40 include realistic regulatory event simulation with 5 event templates (IRB, safety, consent, audit, FDA), patient engagement simulator tracking willingness/fatigue/engagement, site-level negotiation protocol with market state, before/after trajectory visualization, reward-curve dashboard, hypothesis accuracy metric, curriculum injection logging with export, multi-seed reproducibility report with bootstrap CIs, baseline comparison table, and feature/model ablation study table.

A.5 Tier 5: Frontier Research Features

Features 41–50 include multi-agent hierarchical oversight with monitor/reviewer/approver hierarchy, carbon-aware scaling with regional CO2 intensities, federated privacy with differential privacy (Laplace noise, $\epsilon = 1.0$), counterfactual branch-rollout simulation with regret calculation, human-in-the-loop preference alignment using Bradley-Terry model, automated goal discovery from experiment summaries, evolving skill library with success-rate pruning, long-horizon uncertainty quantification with epistemic/aleatoric decomposition, cross-domain transfer (clinical $\rightarrow$ marketing), and NeurIPS-ready reproducibility package with significance testing.

B Environment Configuration

Table 10: Complete environment configuration for each task variant.

Parameter	easy_bench	medium_bench	hard_bench
Max steps	180	180	180
Sites	1	3	5
Budget ($)	120,000	150,000	100,000
Target enrollment	80	120	150
Screening cost	$800	$800	$800
Enrollment cost	$1,200	$1,200	$1,200
Dropout base rate	0.10	0.10	0.10
Uncertainty growth	0.003	0.003	0.003
Token budget	12,000	12,000	12,000
Patient pool size	200	300	400

C Detailed Per-Seed Results

Table 11: Complete per-seed results with standard deviations within each seed's evaluation episodes.

Agent	Seed	Mean Score	Std Score	Episodes
HCAPO	1	0.2547	0.0593	9
HCAPO	7	0.2240	0.0737	9
HCAPO	42	0.2244	0.0703	9
MiRA	1	0.1996	0.0578	9
MiRA	7	0.2280	0.0752	9
MiRA	42	0.2365	0.0758	9
KLong	1	0.2207	0.0714	9
KLong	7	0.2180	0.0671	9
KLong	42	0.1961	0.0657	9
MemexRL	1	0.2560	0.0681	9
MemexRL	7	0.2191	0.0746	9
MemexRL	42	0.2022	0.0679	9

D Statistical Test Details

Table 12: Complete statistical test results for all 6 pairwise comparisons. $n = 3$ seeds per comparison. Bonferroni-adjusted significance threshold: $\alpha_{\text{adj}} = 0.05/6 = 0.0083$.

Pair	$\Delta\mu$	t	p_t	W	p_W	d
HCAPO–KLong	0.0228	2.672	0.0075	0.0	0.109	1.455
HCAPO–MiRA	0.0130	0.615	0.538	3.0	1.000	0.705
HCAPO–MemexRL	0.0086	1.230	0.219	1.0	0.285	0.375
MiRA–KLong	0.0098	0.551	0.581	2.0	0.593	0.587
MemexRL–KLong	0.0142	1.328	0.184	0.0	0.109	0.654
MiRA–MemexRL	-0.0044	-0.162	0.871	3.0	1.000	-0.184

The Wilcoxon signed-rank test does not achieve significance for any comparison due to the small sample size ($n = 3$), which limits its power. With $n = 3$, the minimum achievable Wilcoxon p-value is approximately 0.109, making it impossible to reach $\alpha = 0.05$. The parametric t-test, while making distributional assumptions, provides adequate power for the HCAPO vs. KLong comparison.